SC7 Bayes Methods MT25

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Lecture 8: Estimation of marginal likelihoods,
the Laplace Approximation and BIC

Estimating marginal likelihoods

We have Monte-Carlo tools summarising

$$\pi(\theta|y,m) = \pi(\theta|m)p(y|\theta,m)/p(y|m),$$

the posterior under model m with $\theta \in \Omega_m$. How do we use the MCMC output to do model selection?

Let $\hat{p}_m$ estimate $p(y|m)$ and

$$\hat{B}_{m,m'} = \hat{p}_m/\hat{p}_{m'}$$

estimate Bayes factor comparing models $M = m$ and $M = m'$.

Here are some consistent ML-estimators, of increasing stability.*

*For Laplace estimator see end of lecture. It is biased but consistent as the data size grows. Also Simon Wood, Core Statistics, page 153.

The naive estimate is unbiased and consistent for $p(y|m)$:

1) Simulate $\theta^{(t)} \sim \pi(\theta|m), t = 1...T$ (prior not posterior);

2) set $\hat{p}_m = T^{-1} \sum_t p(y|\theta^{(t)}, m)$.

However, the proportion of samples with large likelihood may be small $\Rightarrow$ high variance.

Harmonic Mean estimate: importance sampling from posterior. Simulate $\theta^{(t)} \sim \pi(\theta|y, m), t = 1...T$. Since

$$E_{\theta^{(t)}|y,m}\left(1/p(y|\theta^{(t)}, m)\right) = 1/p(y|m)$$

the "self-normalised" IS-estimator

$$\hat{p}_m = \left[\frac{1}{T}\sum_t p(y|\theta^{(t)}, m)^{-1}\right]^{-1}$$

is consistent for $p(y|m)$ (if $\theta^{(1:T)}$ eg satisfy ergodic theorem).

Popular but unreliable: rare very large weights when $\theta^{(t)}$ is in the tail of the posterior, so $p(y|\theta^{(t)}, m)$ is very small.

Bridge Estimate: drop model indicator m, so $p(y)$ is $p(y|m)$ etc.

Let $h : \Omega \to R$ be a given function with the property that following expectations are finite and non-zero. The identity

$$p(y) = \frac{E_\theta(\pi(\theta)p(y|\theta)h(\theta))}{E_{\theta|y}(\pi(\theta)h(\theta))}$$

holds. Numerator/denominator expectations over prior/posterior.

Let $\theta^{(1,t)} \sim \pi(\theta)$ and $\theta^{(2,t)} \sim \pi(\theta|y), t = 1...T$ be T be ergodic samples from prior and posterior. The estimator

$$\hat{p} = \frac{\sum_t \pi(\theta^{(1,t)})p(y|\theta^{(1,t)})h(\theta^{(1,t)})}{\sum_t \pi(\theta^{(2,t)})h(\theta^{(2,t)})}.$$

is consistent for $p(y)$.

Choose $h(\theta)$ (see PS2) to make the Relative Mean Square Error

$$RMSE(h) = E((\hat{p} - p(y))^2)/p(y)^2$$

small (expectation over Monte-Carlo sampling variation of $\hat{p}$).

The choice $h(\theta) = \pi(\theta)^{-1} p(y|\theta)^{-1/2}$ is often near optimal so

$$\hat{p} = \frac{\sum_t p(y|\theta^{(1,t)})^{1/2}}{\sum_t p(y|\theta^{(2,t)})^{-1/2}}.$$

This has much lower RMSE than the harmonic mean (see eg RCD example) and is a useful generic estimator.

In order to estimate the Bayes factor we estimate the ML for each model and compute $\hat{B}_{m,m'} = \hat{p}_m / \hat{p}_{m'}$.

If $\Omega_1 = \Omega_2 = \Omega$ (same parameter spaces) then

$$\frac{p(y|m)}{p(y|m')} = \frac{E_{\theta|y,m'}(\pi(\theta|m)p(y|\theta,m)h(\theta))}{E_{\theta|y,m}(\pi(\theta|m')p(y|\theta,m')h(\theta))}$$

If $m = 1$ and $m' = 2$, $\theta^{(1,t)} \sim \pi(\theta|y, m = 1)$ and $\theta^{(2,t)} \sim \pi(\theta|y, m = 2)$, $\quad t = 1...T$, and

$$h(\theta) = (\pi(\theta|m)p(y|\theta,m)\pi(\theta|m')p(y|\theta,m'))^{-1/2},$$

we get

$$\hat{B}_{m,m'} = \frac{\sum_t \left(\frac{\pi(\theta^{(2,t)}|m)p(y|\theta^{(2,t)}|m)}{\pi(\theta^{(2,t)}|m')p(y|\theta^{(2,t)}|m')}\right)^{1/2}}{\sum_t \left(\frac{\pi(\theta^{(1,t)}|m')p(y|\theta^{(1,t)}|m')}{\pi(\theta^{(1,t)}|m)p(y|\theta^{(1,t)}|m)}\right)^{1/2}},$$

convenient, near state of the art for simple generic estimators.

Example: Logistic v. Probit in a Bernoulli GLM

Challenger O-ring data, data $(y_i, x_i), i = 1, 2, ..., n$ with $x_i \in R$ centred and scaled temperature $\texttt{temp}[\texttt{i}]$ and $y_i = 1/0$ if O-ring fails/doesnt fail.

Vector $\beta = (\beta_1, \beta_2)^T$ of effects, linear predictor $\eta_i = (1, x_i)\beta$, observations $y_i \sim Bern(\mu_m(\eta_i))$ in model $m \in \mathcal{M} = \{1, 2\}$,

$$p(y|\theta, m) = \prod_{i=1}^{n} \mu_m(\beta_1 + \beta_2 x_i)^{y_i}(1 - \mu_m(\beta_1 + \beta_2 x_i))^{1-y_i}.$$

Link functions:

1. logistic $\mu_1(\eta) = \exp(\eta)/(1 + \exp(\eta))$;
2. probit $\mu_2(\eta) = \Phi(\eta)$.

Common prior $\beta_1, \beta_2 \sim N(0, 3^2)$ so marginal likelihoods are

$$p(y|m) \propto \int \prod_{i=1}^{n} p(y_i|\beta, m) \exp(-\beta^T \beta/18) d\beta_1 d\beta_2.$$

The estimators above are computed from samples generated by MCMC in the online example code for this lecture.

The resulting estimates of Bayes Factors: 2.7 (naive - pretty good!) 1.9 (harmonic) 2.8 (bridge) 2.7 (R-built-in Laplace-approximation estimator).

Report $\hat{B}_{1,2} = 2.75$, weak evidence "barely worth mentioning" for Logit over Probit.

Laplace Approximation See CR-TBC

Suppose we have an integral I_n to do of the form

$$I_n = \int e^{-nh(\theta)} d\theta$$

Let $\hat{\theta}$ satisfy $h^{(1)}(\hat{\theta}) = 0$ with $h^{(k)}$ the k'th derivative, $\hat{h}^{(k)} = h^{(k)}(\hat{\theta})$ and $\sigma^2 = 1/\hat{h}^{(2)}$ ($\hat{\theta}$ a minimum). It may be shown that

$$I_n = e^{-nh(\hat{\theta})} \sqrt{\frac{2\pi\sigma^2}{n}} \left(1 + \frac{5[\hat{h}^{(3)}]^2\sigma^6 - 3\hat{h}^{(4)}\sigma^4}{24n} + O(n^{-2}) \right)$$

Verified in notes; expand h as a Taylor series to fourth order about $\hat{\theta}$, pull out the constant and quadratic terms, then further expand the exponential $\exp(f(\theta)) = 1 + f + f^2/2 + \ldots$ for the terms involving $\hat{h}^{(3)}$ and $\hat{h}^{(4)}$. This leaves normal moments $E_\theta((\theta - \hat{\theta})^k)$ for $k = 4, 6, 8$.

The multivariate version $\theta = (\theta_1, ..., \theta_p)$ of this is

$$\int e^{-nh(\theta)} d\theta = e^{-nh(\hat{\theta})} (2\pi/n)^{p/2} |\Sigma|^{1/2} (1 + O(1/n))$$

where

$$\Sigma^{-1} = \left. \frac{\partial^2 h}{\partial \theta \partial \theta^T} \right|_{\theta = \hat{\theta}}$$

is the Hessian of $h(\theta)$ at $\hat{\theta}$.

Marginal Likelihood

For a model m and n-component data y

$$p(y|m) = \int_{-\infty}^{\infty} \exp(-n[-\bar{\ell}(\theta) - \log(\pi(\theta))/n])d\theta.$$

where

$$\bar{\ell}(\theta) = \frac{1}{n}\ell(\theta; y).$$

Using the Laplace approximation, with $h(\theta) = -\bar{\ell}(\theta) - \log(\pi(\theta))/n$,

$$\int e^{-nh(\theta)}d\theta = e^{-nh(\hat{\theta})}(2\pi/n)^{p/2}|\Sigma|^{1/2}\,(1 + O(1/n))$$

$$p(y|m) = L(\hat{\theta}; y)\pi(\hat{\theta})(2\pi/n)^{p/2}|\Sigma|^{1/2}\,(1 + O(1/n))$$

where $\hat{\theta}$ is the MAP estimate for θ.

Here $\Sigma^{-1} \simeq J_0(\hat{\theta})$, the observed unit Fisher information.

$$
\begin{aligned}
\Sigma^{-1} &= \left.\frac{\partial^2 h(\theta)}{\partial\theta\partial\theta^T}\right|_{\theta=\hat{\theta}} \\
&= -\frac{1}{n}\left.\frac{\partial^2 \ell(\theta;y)}{\partial\theta\partial\theta^T}\right|_{\theta=\hat{\theta}} - \frac{1}{n}\left.\frac{\partial^2 \pi(\theta)}{\partial\theta\partial\theta^T}\right|_{\theta=\hat{\theta}} \\
&\simeq J_0
\end{aligned}
$$

if $\ell(\theta;y) = \sum_i \ell(\theta;y_i)$ is $O(n)$ and $\pi(\theta)$ is $O(1)$. This gives

$$\log(p(y|m)) \simeq \ell(\hat{\theta};y) + \log(\pi(\hat{\theta})) + \frac{p}{2}\log(2\pi/n) - \frac{1}{2}\log(|J_0|) \tag{1}$$

$$\simeq \ell(\hat{\theta}_{MLE};y) - \frac{p}{2}\log(n) + O(1) \tag{2}$$

The BIC is -2 times (2). Our estimator is (1).

The BIC drops constant terms not just terms which go to zero!

The model with the least BIC score has the largest marginal likelihood, approximately.

In "The Bayesian Choice", Christian Robert notes that the BIC isn't Bayesian, as the prior doesn't enter. It is widely used in Frequentist analysis alongside the AIC. Expect it to be useful at large n only.

Equation (1) is often reasonably accurate at large n and implemented in many software packages. We used it in the O-rings example above.

Notice that the LA shows the parsimony of model selection with the marginal likelihood - there is a penalty $-p\log(n)$ for model dimension p.